



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BENGALURU – 560100, KARNATAKA, INDIA

B.TECH. (CSE)
V SEMESTER
UE24CS341A – SOFTWARE ENGINEERING

PROJECT TEST PLAN
ON
UpSkill-AI: An AI-Powered Upskilling Recommendation
and Automation System for Students
Phase-3 Deliverable

TEAM NUMBER: 18 PROJECT ID: 22

SUBMITTED BY

Name	SRN	Module Owned
VANSH SHARMA	PES2UG24AM179	M1 – Profile & Skill Extraction
YAJAT SONI	PES2UG24AM185	M2 – Gap Analysis & Roadmap
HEMANTH KUMAR R	PES2UG24AM202	M3 – Progress & Adaptive Adjustment
SHUBHIKA PRADEEP	PES2UG24AM157	M4 – Catalogue, Auth & RBAC

August – November 2026

Revision History

Name	Date	Reason For Changes	Version
VANSH SHARMA, YAJAT SONI, HEMANTH KUMAR R, SHUBHIKA PRADEEP	30-09-2026	Initial version	1.0

1. Introduction

1.1 Purpose

This document defines the test plan for UpSkill-AI. It states what will be tested, at which level, in what environment, and against which criteria, and it contains the full set of test cases derived from the functional requirements in the Software Requirements Specification.

1.2 Current Execution Status

This version of the test plan is a planning deliverable. The test cases have been designed and reviewed against the SRS, but implementation of the system has not yet begun, so none of the cases have been executed.

Accordingly, the Actual Result column of every test case reads "Not executed — pending implementation" and the Test Result column reads "Pending". These two columns will be populated during Phase-3 execution, once the corresponding functionality is implemented, and this document will be reissued at version 2.0 with the observed results and pass or fail verdicts.

No test case in this document has been marked as passed on the basis of expected behaviour alone.

1.3 Scope of Testing

In scope:

- Unit testing of individual functions and service methods within each of the four modules
- Integration testing of the interfaces between modules, and between modules and the database and external service
- System testing of complete user journeys across the four roles
- Security testing of authentication, authorisation boundaries and error disclosure
- Validation of stated performance targets under a representative dataset

Out of scope for this phase:

- Load and stress testing beyond the expected single-cohort workload
- Penetration testing by an external party
- Cross-browser compatibility testing beyond the two most recent major versions of Chrome, Firefox and Edge
- Accuracy benchmarking of resume extraction against a labelled corpus; extraction is validated behaviourally through the student confirmation step

1.4 References

Ref	Document	Version
R1	UpSkill-AI Software Requirements Specification	1.0
R2	UpSkill-AI High Level Software Architecture Document (Team 18)	1.0
R3	Jackfruit Software Engineering Mini-Project Guidelines 2026, UE24CS341A	2026
R4	UpSkill-AI Product Backlog (Team 18)	1.0

2. Test Strategy

2.1 Test Levels and Identifier Convention

Test case identifiers follow the convention recorded in Appendix C of the Software Requirements Specification, so that every requirement traces to a test case that actually exists in this document:

- TC-Fn-nn — unit and integration test cases for feature n
- STC-Fn-nn — system test cases for feature n

The test level of each case (Unit, Integration or System) is stated in the Name of Module column of every test case row, so the type remains visible without overloading the identifier.

Level	What it covers	Who executes	Tooling
Unit	A single function or service method in isolation, with dependencies replaced by fixtures or mocks. No database, no network.	Module owner	pytest, pytest-cov (backend); Jest (frontend)
Integration	Interaction across a boundary — module to module, module to database, or module to external service.	Module owners jointly	pytest with a disposable test database; mocked external client
System	A complete user journey through the deployed stack, exercised as a real user of a given role would.	Whole team	Manual execution against the docker-compose stack

2.2 Test Design Approach

- Every functional requirement in the SRS maps to at least one test case; the mapping is given in §7.
- Each module has at least ten test cases spanning all three levels, satisfying the Phase-3 requirement of 8 to 10 cases per implemented use case.
- Both valid and invalid inputs are covered. Invalid-input cases are written explicitly rather than assumed, because the SRS specifies exact rejection messages for several of them.
- Boundary values are tested where the requirement states a limit — file size at 5 MB, description length at 2000 characters, completion percentage at 0 and 100, confidence threshold at the configured value.
- Security cases test the boundary from the wrong side: each protected operation is attempted with an insufficient role, and the expected result is denial.
- Cases for deterministic logic (gap computation, prerequisite sequencing) assert exact outputs from fixed fixtures, which is possible only because ADR-02 made that logic rule-based rather than learned.

2.3 Test Environment

Component	Configuration
Application stack	docker-compose: upskill-web (Next.js), upskill-api (FastAPI), upskill-db (PostgreSQL 16)
Test database	Disposable PostgreSQL instance, recreated and migrated before each integration test run
External service	YouTube Data API v3 client replaced by a mock in all automated tests; no test consumes live quota
Backend test runner	pytest with pytest-cov; coverage report published by the Jenkins pipeline
Frontend test runner	Jest with React Testing Library
Static analysis	SonarQube quality gate; a failing gate blocks promotion of the build
Browsers for system tests	Chrome, Firefox and Edge — latest two major versions
Test data	Seeded fixture set: 3 student accounts, 1 faculty account, 1 administrator account, a 40-entry skill taxonomy, 2 target roles, 25 catalogue resources

2.4 Entry and Exit Criteria

	Entry criteria	Exit criteria
Unit testing	Module code compiles; dependencies mockable; fixtures prepared	All unit cases for the module executed; no failing case; statement coverage target met for the module
Integration testing	All participating modules pass their unit tests; test database provisioned; shared contracts frozen	All integration cases executed; no failing case; no unresolved contract mismatch between modules
System testing	Full stack deploys from docker-compose; seeded test data loaded; all integration tests passing	All system cases executed; no open defect of severity High or above; all stated performance targets met

2.5 Defect Severity Definitions

Severity	Definition	Target resolution
Critical	Data loss, authentication bypass, or a role boundary that can be crossed	Immediate; blocks all further testing
High	A core user journey cannot be completed; incorrect gap or recommendation output	Before the sprint ends
Medium	A feature behaves incorrectly but a workaround exists	Within the current sprint if capacity allows
Low	Cosmetic, wording, or minor usability issue	Backlog; fixed opportunistically

2.6 Roles and Responsibilities

Member	SRN	Module	Test responsibility
Vansh Sharma	PES2UG24AM179	M1 – Profile & Skill Extraction	Authors and executes TC-F1-01 to TC-F1-15, STC-F1-01 to STC-F1-02
Yajat Soni	PES2UG24AM185	M2 – Gap Analysis & Roadmap	Authors and executes TC-F2-01 to TC-F2-12, STC-F2-01 to STC-F2-02
Hemanth Kumar R	PES2UG24AM202	M3 – Progress & Adaptive Adjustment	Authors and executes TC-F3-01 to TC-F3-12, STC-F3-01 to STC-F3-02
Shubhika Pradeep	PES2UG24AM157	M4 – Catalogue, Auth & RBAC	Authors and executes TC-F4-01 to TC-F4-14, STC-F4-01 to STC-F4-02
All members	—	Cross-module	Joint review of integration cases; system test execution and defect triage

2.7 Testing Risks

#	Risk	Mitigation
T1	Test cases cannot be executed until the corresponding functionality is implemented, compressing execution into a short window near the deadline.	Cases are written per module and executed as each module completes, rather than held to a single end-of-project testing phase.
T2	Integration tests depend on the shared skill-vector and taxonomy contracts. A late change to either invalidates cases across several modules.	Both contracts are frozen before implementation begins (architecture rule R3); any change requires all four owners to agree and re-review affected cases.
T3	Seeded taxonomy and catalogue data is required before most cases can run meaningfully.	Seeding is a Sprint 1 backlog item shared across the team, not left to one owner.
T4	Tests that touch the external API would be flaky and would consume quota.	The external client sits behind the M4 interface and is mocked in every automated test.

3. Test Cases — M1: Student Profile and Skill Extraction

Requirements covered: REQ-F1-01 to REQ-F1-40. Test case IDs: TC-F1-01 to TC-F1-15, STC-F1-01 to STC-F1-02.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F1-01	M1 – Profile Management [Unit]	Verify a student can create exactly one profile linked to their account	Student account exists and is authenticated	1: Log in as student 2: Open profile page 3: Enter profile details 4: Submit	Email: s1@pes.edu Name: Test Student Education level: UG	Profile created and linked to the account; a second create attempt is rejected with "Profile already exists"	Not executed — pending implementation	Pending
TC-F1-02	M1 – Profile Management [Unit]	Verify a course entry is saved with all mandatory fields	Student profile exists	1: Open profile 2: Select Add Course 3: Enter course details 4: Submit	Course: Data Structures Code: UE23CS251A Grade: A Term: Sem 3	Course entry saved and displayed in the course list	Not executed — pending implementation	Pending
TC-F1-03	M1 – Profile Management [Unit]	Verify an invalid grade value is rejected	Student profile exists	1: Open Add Course 2: Enter a grade outside the permitted set 3: Submit	Course: DBMS Grade: Z Term: Sem 4	Save rejected; message identifies grade as the invalid field; no record written	Not executed — pending implementation	Pending
TC-F1-04	M1 – Profile Management [Unit]	Verify a project description exceeding 2000 characters is rejected	Student profile exists	1: Open Add Project 2: Paste an over-length description 3: Observe the character counter 4: Submit	Title: Test Project Description: 2050 characters	Save rejected; current character count displayed; no record written	Not executed — pending implementation	Pending
TC-F1-05	M1 – Resume Upload [Unit]	Verify only PDF and DOCX resume uploads are accepted	Student is logged in	1: Open resume upload 2: Select a .txt file 3: Submit	resume_sample.txt (12 KB)	Upload rejected with "Only PDF and DOCX files are accepted."	Not executed — pending implementation	Pending
TC-F1-06	M1 – Resume Upload [Unit]	Verify a resume file larger than 5 MB is rejected	Student is logged in	1: Open resume upload 2: Select an oversized PDF 3: Submit	large_resume.pdf (6.2 MB)	Upload rejected with "File exceeds the 5 MB limit."	Not executed — pending implementation	Pending
TC-F1-07	M1 – Resume Upload [Unit]	Verify an unreadable resume yields a manual-entry fallback rather than an error	Student is logged in	1: Upload a scanned image-only PDF 2: Wait for processing to finish	scanned_only.pdf (text layer under 100 characters)	Student is informed the resume could not be read and is offered manual skill entry; profile unchanged	Not executed — pending implementation	Pending
TC-F1-08	M1 – Skill Normalisation [Unit]	Verify known variant spellings resolve to one canonical skill	Skill taxonomy is seeded with the React canonical skill and its aliases	1: Submit each variant to the skill resolver 2: Compare the returned canonical id for each	"ReactJS", "React.js", "react", " REACT "	All four variants resolve to the same canonical skill "React"; no duplicate skill entries created	Not executed — pending implementation	Pending
TC-F1-09	M1 – Skill Normalisation [Unit]	Verify an unresolvable skill string is retained for review and not discarded	Skill taxonomy is seeded	1: Submit an unknown skill string 2: Inspect the unresolved list	"Quantum Blockchain Fabric"	String recorded as unresolved and presented to the student for manual mapping; it is not silently dropped	Not executed — pending implementation	Pending
TC-F1-10	M1 – Skill Vector [Unit]	Verify the highest level is used when a skill is derived from two sources	Student has the same skill from a course and from a resume at different levels	1: Add skill at Beginner from a course 2: Confirm the same skill at Advanced from a resume 3: Read the skill vector	Skill: Python Source A level: Beginner Source B level: Advanced	Skill vector contains one Python entry at Advanced; both contributing sources recorded	Not executed — pending implementation	Pending
TC-F1-11	M1 – Skill Vector [Unit]	Verify an incoming level lower than the held level does not lower the stored	Student holds Python at Advanced	1: Call apply_vector_update with a lower level 2: Read the skill vector	Skill: Python Incoming level: Intermediate	Stored level remains Advanced; the call is a no-op and does not raise	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		level						
TC-F1-12	M1 – Target Role [Unit]	Verify only one active target role can be held at a time	Student profile exists; two roles are defined	1: Select target role A 2: Select target role B 3: Read the active role	Role A: Backend Developer Role B: Data Analyst	Only Data Analyst is active; existing recommendations are marked stale on the change	Not executed — pending implementation	Pending
TC-F1-13	M1 ↔ M2 Integration [Integration]	Verify the skill vector produced by M1 is consumed correctly by gap analysis	Profile with known skills exists; target role seeded with required skills	1: Build the profile 2: Select target role 3: Request the gap from M2 4: Compare against the expected gap	Held: Python (Intermediate), SQL (Beginner) Role requires: Python (Advanced), SQL (Intermediate), Docker (Beginner)	Gap returns Python (under-level), SQL (under-level) and Docker (absent) — three entries, computed level-aware	Not executed — pending implementation	Pending
TC-F1-14	M1 ↔ M3 Integration [Integration]	Verify a completion event raises the skill level through the correct owning module	Student enrolled in a resource teaching Docker at Beginner	1: Mark the resource complete with valid evidence 2: Read the skill vector from M1	Resource teaches: Docker (Beginner) Held before: none	Docker appears in the skill vector at Beginner with source "completion"; change recorded in skill_change_history	Not executed — pending implementation	Pending
TC-F1-15	M1 – Access Control [Integration]	Verify a student cannot read another student's profile	Two student accounts exist	1: Log in as student A 2: Request student B's profile by id	Actor: student A Target: student B profile id	Request denied with an authorisation error; no profile data returned	Not executed — pending implementation	Pending
STC-F1-01	M1 – System [System]	End-to-end: new student builds a profile and reaches a computed gap	System deployed; taxonomy and one target role seeded	1: Register 2: Log in 3: Add two courses and one project 4: Upload a resume and confirm extracted skills 5: Select a target role 6: View the gap summary	Full valid student dataset	Gap summary displays required skill count, held skill count and gap count consistent with the entered profile	Not executed — pending implementation	Pending
STC-F1-02	M1 – System [System]	End-to-end: profile correction propagates to recommendations	Student has a generated roadmap	1: Remove a skill from the profile 2: Return to the roadmap screen	Remove skill: SQL	Recommendations are marked stale and the refreshed roadmap includes SQL-closing resources	Not executed — pending implementation	Pending

4. Test Cases — M2: Skill Gap Analysis and Roadmap Generation

Requirements covered: REQ-F2-01 to REQ-F2-11. Test case IDs: TC-F2-01 to TC-F2-12, STC-F2-01 to STC-F2-02.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F2-01	M2 – Gap Analysis [Unit]	Verify gap computation treats an under-level skill as a gap	Target role seeded with required levels	1: Supply a skill vector holding a skill below the required level 2: Run compute_gap	Held: Java (Beginner) Required: Java (Advanced)	Java is returned as a gap entry with current and required levels shown	Not executed — pending implementation	Pending
TC-F2-02	M2 – Gap Analysis [Unit]	Verify a skill held at or above the required level is not reported as a gap	Target role seeded	1: Supply a vector meeting all requirements 2: Run compute_gap	Held: Java (Advanced) Required: Java (Intermediate)	Java is absent from the gap; gap count for that skill is zero	Not executed — pending implementation	Pending
TC-F2-03	M2 – Gap Analysis [Unit]	Verify gap computation is a pure function with no external dependency	Test fixtures only; database and network disabled	1: Call compute_gap with fixture inputs twice 2: Compare outputs	Fixed fixture vector and role	Identical output on both calls; no database or network access attempted	Not executed — pending implementation	Pending
TC-F2-04	M2 – Ranking [Unit]	Verify gap entries are ranked by relevance to the target role	Role seeded with weighted skill requirements	1: Compute the gap for a multi-skill shortfall 2: Inspect the returned order	Gap: Docker, Java, Git with differing role weights	Entries returned in descending relevance order, highest-weighted skill first	Not executed — pending implementation	Pending
TC-F2-05	M2 – Sequencing [Unit]	Verify a prerequisite resource is ordered before the resource that depends on it	Taxonomy contains a prerequisite edge	1: Build a roadmap covering both skills 2: Inspect item order	Prerequisite: SQL → Advanced SQL Tuning	The SQL resource appears before the Advanced SQL Tuning resource in the roadmap	Not executed — pending implementation	Pending
TC-F2-06	M2 – Sequencing [Unit]	Verify a cyclic prerequisite definition is detected and does not cause a hang	Taxonomy seeded with a deliberate cycle	1: Request a roadmap covering the cyclic skills 2: Observe the response	Cycle: Skill X → Skill Y → Skill X	Cycle detected and reported; request terminates with a clear error rather than looping indefinitely	Not executed — pending implementation	Pending
TC-F2-07	M2 – Explainability [Unit]	Verify every roadmap item records the specific gap it closes	Roadmap generated for a known gap	1: Generate a roadmap 2: Inspect each item's linked gap field	Gap: Docker (absent), Java (under-level)	Each returned item carries a non-empty reference to the gap skill it addresses	Not executed — pending implementation	Pending
TC-F2-08	M2 – Staleness [Unit]	Verify existing recommendations are marked stale when the target role changes	Student has a generated roadmap	1: Change the target role 2: Inspect the recommendation status	New role: Data Analyst	All existing recommendation records are flagged stale	Not executed — pending implementation	Pending
TC-F2-09	M2 – Caching [Unit]	Verify a repeated external lookup for the same skill uses the cache	Cache enabled; external client mocked with a call counter	1: Request resources for a skill twice within the TTL 2: Read the mock call count	Skill: Kubernetes TTL: 24 hours	External client called exactly once; second request served from cache	Not executed — pending implementation	Pending
TC-F2-10	M2 ↔ M4 Integration [Integration]	Verify only approved catalogue resources reach a student roadmap	Catalogue contains approved and pending resources for the same skill	1: Generate a roadmap covering that skill 2: Inspect returned resources	Skill: Docker Resources: 2 approved, 1 pending	Only the two approved resources appear; the pending resource is excluded	Not executed — pending implementation	Pending
TC-F2-11	M2 ↔ M4 Integration [Integration]	Verify fallback behaviour when the catalogue has no resource for a gap skill	Catalogue deliberately empty for the target skill; external client mocked	1: Generate a roadmap covering the uncovered skill 2: Observe the response	Skill: Rust (no catalogue entries)	External lookup or cached fallback list is returned; no error is shown to the student and the rest of the roadmap is unaffected	Not executed — pending implementation	Pending
TC-F2-12	M2 ↔ M4 Integration	Verify external service failure degrades	Catalogue empty for the skill;	1: Generate a roadmap 2: Observe the response and logs	Skill: Rust Mock: connection	Roadmap returns successfully with the remaining items;	Not executed — pending	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
	[Integration]	gracefully	external client mocked to raise a timeout		timeout	failure logged with correlation id; student sees a plain-language notice, not a stack trace	implementation	
STC-F2-01	M2 – System [System]	End-to-end: student receives an ordered, explainable roadmap	Profile, target role, seeded taxonomy and catalogue all present	1: Log in 2: Open the roadmap screen 3: Expand each gap skill	Complete valid dataset	Roadmap displays prioritised gaps, ordered resources respecting prerequisites, and the gap each resource closes	Not executed — pending implementation	Pending
STC-F2-02	M2 – System [System]	Verify roadmap response time meets the stated performance target	System deployed with a representative dataset	1: Request a roadmap for a typical profile 2: Record elapsed time over 10 runs	Profile with 15 held skills; role requiring 20 skills	Roadmap returned within 5 seconds on each run (NFR-PERF-03)	Not executed — pending implementation	Pending

5. Test Cases — M3: Progress Tracking and Adaptive Adjustment

Requirements covered: REQ-F3-01 to REQ-F3-17. Test case IDs: TC-F3-01 to TC-F3-12, STC-F3-01 to STC-F3-02.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F3-01	M3 – Enrolment [Unit]	Verify a student can enrol in a recommended resource	Roadmap generated for the student	1: Open the roadmap 2: Select a resource 3: Click Enrol	Resource id: LI-1001	Progress record created with status "enrolled" and completion 0%	Not executed — pending implementation	Pending
TC-F3-02	M3 – Enrolment [Unit]	Verify enrolment is rejected for a non-existent resource	Student is authenticated	1: Submit an enrolment request with an invalid resource id	Resource id: LI-999999	Request rejected with a clear error; no progress record created	Not executed — pending implementation	Pending
TC-F3-03	M3 – Progress Update [Unit]	Verify completion percentage is accepted within the valid range	Student is enrolled in a resource	1: Update progress to a valid value 2: Read the stored record	Completion: 60	Progress stored as 60% with status "in-progress"	Not executed — pending implementation	Pending
TC-F3-04	M3 – Progress Update [Unit]	Verify an out-of-range completion percentage is rejected	Student is enrolled in a resource	1: Submit an invalid completion value	Completion: 140	Update rejected with a validation message; stored value unchanged	Not executed — pending implementation	Pending
TC-F3-05	M3 – Completion [Unit]	Verify a completion event against a non-enrolled resource is blocked	Student is not enrolled in the target resource	1: Submit a completion request for that resource	Resource id: LI-1002 (not enrolled)	Completion rejected; no progress or evidence record created	Not executed — pending implementation	Pending
TC-F3-06	M3 – Evidence [Unit]	Verify required completion evidence is validated before completion is accepted	Resource configured to require evidence	1: Mark the resource complete with no evidence attached 2: Observe the response	Evidence: none	Completion not accepted; student is prompted to supply valid evidence	Not executed — pending implementation	Pending
TC-F3-07	M3 – Evidence [Unit]	Verify a self-assessment score outside the valid range is rejected	Student enrolled; quiz evidence enabled	1: Submit an invalid quiz score	Score: 130	Score rejected with a validation message; no evidence record written	Not executed — pending implementation	Pending
TC-F3-08	M3 – Stalled Path [Unit]	Verify a learning path with no qualifying progress is identified as stalled	Enrolment exists with last progress older than the configured interval	1: Run the stalled-path detection job 2: Inspect flagged paths	Last progress: 21 days ago Configured interval: 14 days	The path is flagged as stalled and a reminder is scheduled	Not executed — pending implementation	Pending
TC-F3-09	M3 – Progress History [Unit]	Verify progress history is preserved when recommendations are re-ranked	Student has progress records and a stale roadmap	1: Trigger a re-rank 2: Read the progress records	Existing progress on two resources	Both progress records remain intact with their original values after re-ranking	Not executed — pending implementation	Pending
TC-F3-10	M3 ↔ M1 Integration [Integration]	Verify validated evidence updates the skill vector through M1 only	Student enrolled in a resource teaching a new skill	1: Complete the resource with valid evidence 2: Inspect the skill vector and its change history	Resource teaches: Git (Intermediate)	Skill vector updated via M1; change history records previous level, new level and source "completion"	Not executed — pending implementation	Pending
TC-F3-11	M3 ↔ M2 Integration [Integration]	Verify a skill-vector change triggers re-ranking rather than waiting for next login	Student has an active roadmap	1: Complete a resource with valid evidence 2: Without logging out, request the roadmap	Resource closes the Docker gap	Roadmap is recomputed; the closed Docker gap no longer appears as an outstanding item	Not executed — pending implementation	Pending
TC-F3-12	M3 ↔ M4 Integration [Integration]	Verify a faculty mentor can view an assigned mentee's progress but cannot modify it	Faculty account with one assigned mentee	1: Log in as faculty 2: Open the mentee's progress 3: Attempt to edit a progress value	Faculty: F-01 Mentee: S-01	Progress is visible; any modification attempt is denied with an authorisation error	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
STC-F3-01	M3 – System [System]	End-to-end: completing a resource closes a gap and updates the dashboard	Full system deployed with a generated roadmap	1: Enrol in a resource 2: Update progress to 100% 3: Upload completion evidence 4: Open the progress dashboard	Valid certificate file (PDF, 400 KB)	Dashboard shows the gap as closed, the milestone recorded, and the roadmap refreshed	Not executed — pending implementation	Pending
STC-F3-02	M3 – System [System]	Verify a stalled learner receives a reminder notification	Enrolment left inactive beyond the configured interval	1: Advance the system clock or seed an old progress timestamp 2: Run the scheduled detection job 3: Check the student's notifications	Last activity: 21 days ago	A stalled-path reminder is generated and visible to the student	Not executed — pending implementation	Pending

6. Test Cases — M4: Resource Catalogue, Authentication and Access Control

Requirements covered: REQ-F4-01 to REQ-F4-18, NFR-SEC-01 to NFR-SEC-10. Test case IDs: TC-F4-01 to TC-F4-14, STC-F4-01 to STC-F4-02.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F4-01	M4 – Authentication [Unit]	Verify a valid login issues a token and grants access	Registered student account exists	1: Submit valid credentials to the login endpoint 2: Use the returned token on a protected route	Email: s1@pes.edu Password: correct value	Token issued; protected route returns 200 with the student's data	Not executed — pending implementation	Pending
TC-F4-02	M4 – Authentication [Unit]	Verify an invalid password is rejected without revealing which field was wrong	Registered account exists	1: Submit a valid email with an incorrect password	Email: s1@pes.edu Password: wrong value	Login rejected with a generic credentials error; no token issued; no indication of which field failed	Not executed — pending implementation	Pending
TC-F4-03	M4 – Authentication [Unit]	Verify stored passwords are never held in plaintext	Account registration completed	1: Register a new account 2: Inspect the stored user record	Password: TestPass#2026	Stored value is an Argon2id hash; the plaintext password does not appear in the record or in logs	Not executed — pending implementation	Pending
TC-F4-04	M4 – Authorisation [Unit]	Verify a student cannot access an administrator-only endpoint	Student account authenticated	1: Call the catalogue creation endpoint with a student token	Actor role: Student Target: POST /catalogue	Request denied with an authorisation error; no resource created	Not executed — pending implementation	Pending
TC-F4-05	M4 – Authorisation [Unit]	Verify a faculty mentor cannot perform administrator catalogue management	Faculty account authenticated	1: Call the catalogue moderation endpoint with a faculty token	Actor role: Faculty-Mentor Target: PATCH /catalogue/{id}/moderate	Request denied with an authorisation error; moderation state unchanged	Not executed — pending implementation	Pending
TC-F4-06	M4 – Authorisation [Unit]	Verify a faculty mentor cannot view a student who is not their assigned mentee	Faculty account with one assigned mentee; a second unassigned student exists	1: Request the unassigned student's progress	Faculty: F-01 Target: S-02 (unassigned)	Request denied; no student data returned	Not executed — pending implementation	Pending
TC-F4-07	M4 – Catalogue [Unit]	Verify a resource cannot be published without required metadata	Administrator authenticated	1: Submit a resource with no skill tags 2: Observe the response	Title: Intro to Go Skill tags: none	Creation rejected; validation message identifies the missing mandatory field	Not executed — pending implementation	Pending
TC-F4-08	M4 – Catalogue [Unit]	Verify a resource skill tag must correspond to a canonical skill	Administrator authenticated; taxonomy seeded	1: Submit a resource tagged with an unknown skill	Skill tag: "Blockchain Quantum"	Creation rejected; administrator is shown the closest matching canonical skills	Not executed — pending implementation	Pending
TC-F4-09	M4 – Catalogue [Unit]	Verify catalogue search returns matches on title, description and skill tags	Catalogue seeded with known resources	1: Search using a term present only in a skill tag 2: Inspect results	Search term: "Docker"	All resources tagged with Docker are returned; unrelated resources are excluded	Not executed — pending implementation	Pending
TC-F4-10	M4 – Catalogue [Unit]	Verify an inactive resource does not appear in normal student search results	Catalogue contains one resource marked inactive	1: Search as a student for that resource's title	Resource state: inactive	Inactive resource is not returned in student-facing results	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F4-11	M4 – Audit [Unit]	Verify a faculty override is recorded with an attributable audit entry	Faculty account with an assigned mentee and an active recommendation	1: Override a recommendation with a stated reason 2: Inspect the audit record	Reason: "Student already completed equivalent coursework"	Audit record written with actor, action, target and reason; the original algorithmic recommendation is retained, not deleted	Not executed — pending implementation	Pending
TC-F4-12	M4 – Integration [Integration]	Verify authorisation is enforced server-side and not only in the interface	Student account authenticated	1: Call an administrator endpoint directly, bypassing the web interface	Direct API call with a student token	Request denied by the server regardless of interface state (NFR-SEC-02)	Not executed — pending implementation	Pending
TC-F4-13	M4 – Integration [Integration]	Verify error responses do not leak internal implementation details	System deployed	1: Trigger an internal error on a protected route 2: Inspect the response body	Malformed request causing a handler exception	Generic error returned with a correlation id; no stack trace, token or internal identifier present (NFR-SEC-10)	Not executed — pending implementation	Pending
TC-F4-14	M4 – Integration [Integration]	Verify the database is not reachable from outside the container network	System running under docker-compose	1: Attempt a direct database connection from the host	Host connection to port 5432	Connection refused; database is reachable only from inside the shared Docker network	Not executed — pending implementation	Pending
STC-F4-01	M4 – System [System]	End-to-end: administrator seeds a resource that reaches a student roadmap	System deployed; taxonomy seeded	1: Log in as administrator 2: Create and approve a resource for a known gap skill 3: Log in as a student with that gap 4: Open the roadmap	Resource: "Docker Fundamentals", tagged Docker, Beginner, approved	The newly approved resource appears in the student's roadmap against the Docker gap	Not executed — pending implementation	Pending
STC-F4-02	M4 – System [System]	End-to-end: faculty endorsement is visible to the student	Faculty assigned to a student who has an active roadmap	1: Log in as faculty 2: Endorse one recommendation 3: Log in as the student 4: Open the roadmap	Recommendation id: R-2001	The endorsement is shown against that recommendation in the student's view and recorded in the audit log	Not executed — pending implementation	Pending

7. Requirement to Test Case Traceability

Every functional requirement group maps to at least one test case. This table supplies the Test Case ID and System Test Case ID columns of Appendix C of the SRS, using the identifiers recorded there.

Requirement IDs	Requirement group	Unit / Integration cases	System cases
REQ-F1-01 to F1-07	Profile management	TC-F1-01, TC-F1-02, TC-F1-03, TC-F1-04	STC-F1-01
REQ-F1-08 to F1-18	Resume upload and extraction	TC-F1-05, TC-F1-06, TC-F1-07	STC-F1-01
REQ-F1-19 to F1-24	Skill normalisation	TC-F1-08, TC-F1-09	STC-F1-01
REQ-F1-25 to F1-30	Skill vector computation	TC-F1-10, TC-F1-11, TC-F1-13, TC-F1-14	STC-F1-02
REQ-F1-31 to F1-34	Target role selection	TC-F1-12	STC-F1-01
REQ-F1-35 to F1-38	Validation and error handling	TC-F1-03, TC-F1-04, TC-F1-07	STC-F1-01
REQ-F1-39, F1-40	Profile access control	TC-F1-15	STC-F4-02
REQ-F2-01 to F2-04	Gap computation and ranking	TC-F2-01, TC-F2-02, TC-F2-03, TC-F2-04, TC-F1-13	STC-F2-01
REQ-F2-05 to F2-08	Roadmap generation and sequencing	TC-F2-05, TC-F2-06, TC-F2-07, TC-F2-08, TC-F2-10	STC-F2-01
REQ-F2-09, F2-10	External fallback and caching	TC-F2-09, TC-F2-11, TC-F2-12	STC-F2-01
REQ-F2-11	Recalculation timestamp	TC-F2-08	STC-F1-02
REQ-F3-01 to F3-04	Enrolment and progress recording	TC-F3-01, TC-F3-02, TC-F3-03, TC-F3-04	STC-F3-01
REQ-F3-05 to F3-07	Completion evidence	TC-F3-05, TC-F3-06, TC-F3-07	STC-F3-01
REQ-F3-08, F3-09	Adaptive vector update and re-rank	TC-F3-10, TC-F3-11	STC-F3-01
REQ-F3-10 to F3-12	Dashboards and milestones	TC-F3-09	STC-F3-01
REQ-F3-13, F3-14	Stalled paths and reminders	TC-F3-08	STC-F3-02
REQ-F3-15 to F3-17	History preservation and validity	TC-F3-05, TC-F3-09	STC-F3-01
REQ-F4-01, F4-02	Authentication and role assignment	TC-F4-01, TC-F4-02, TC-F4-03	STC-F4-01
REQ-F4-03, F4-04, F4-18	Authorisation enforcement	TC-F4-04, TC-F4-05, TC-F4-12	STC-F4-02
REQ-F4-05 to F4-08	Catalogue creation and validation	TC-F4-07, TC-F4-08	STC-F4-01
REQ-F4-09 to F4-12	Catalogue search and moderation	TC-F4-09, TC-F4-10, TC-F2-10	STC-F4-01
REQ-F4-13 to F4-15	Faculty views, endorse and override	TC-F4-06, TC-F3-12	STC-F4-02
REQ-F4-16	Audit records	TC-F4-11	STC-F4-02
NFR-PERF-03	Recommendation response time	—	STC-F2-02
NFR-SEC-02, SEC-08	Server-side authorisation and mentee scoping	TC-F4-06, TC-F4-12	STC-F4-02
NFR-SEC-03	Password hashing	TC-F4-03	—
NFR-SEC-10	Error disclosure	TC-F4-13	—
NFR-SEC-01	Network isolation of the database	TC-F4-14	—

8. Test Summary

Module	Unit	Integration	System	Total
M1 – Profile & Skill Extraction	12	3	2	17
M2 – Gap Analysis & Roadmap	9	3	2	14
M3 – Progress & Adaptive Adjustment	9	3	2	14
M4 – Catalogue, Auth & RBAC	11	3	2	16
Total	41	12	8	61

Every module exceeds the Phase-3 minimum of 8 to 10 test cases per implemented use case, and every module has coverage at all three test levels.

8.1 Execution Record

To be completed during Phase-3 execution and reissued at version 2.0.

Metric	Target	Observed
Test cases executed	61 of 61	To be recorded
Test cases passed	61	To be recorded
Test cases failed	0	To be recorded
Defects raised (Critical / High)	0 open at exit	To be recorded
Backend statement coverage	Per-module target met	To be recorded
SonarQube quality gate	Passed	To be recorded